

Лабораторная работа №15

Динамическая маршрутизация

Поляков Г. С.

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Российский университет дружбы народов, Москва, Россия

Информация

- Поляков Глеб Сергеевич
- студент
- Российский университет дружбы народов

Выполнение лабораторной работы

```
msk-donskaya-gspolyakov-gw-1(config)#router ospf 1
msk-donskaya-gspolyakov-gw-1(config-router)#rou
msk-donskaya-gspolyakov-gw-1(config-router)#router-id 10.128.254.1
msk-donskaya-gspolyakov-gw-1(config-router)#netw
msk-donskaya-gspolyakov-gw-1(config-router)#network 10.0.0.0 0.255.255.255 ar
msk-donskaya-gspolyakov-gw-1(config-router)#network 10.0.0.0 0.255.255.255 area 0
msk-donskaya-gspolyakov-gw-1(config-router)#exit
```

```
msk-donskaya-gspolyakov-gw-1#sh ip ospf
Routing Process "ospf 1" with ID 10.128.254.1
Supports only single TOS(TOS0) routes
Supports opaque LSA
SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
Number of external LSA 0. Checksum Sum 0x000000
Number of opaque AS LSA 0. Checksum Sum 0x000000
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
External flood list length 0
  Area BACKBONE(0)
    Number of interfaces in this area is 7
    Area has no authentication
    SPF algorithm executed 1 times
    Area ranges are
    Number of LSA 1. Checksum Sum 0x007286
    Number of opaque link LSA 0. Checksum Sum 0x000000
    Number of DCbitless LSA 0
    Number of indication LSA 0
    Number of DoNotAge LSA 0
    Flood list length 0

msk-donskaya-gspolyakov-gw-1#sh ip ospf ne
msk-donskaya-gspolyakov-gw-1#sh ip ospf neighbor
```

```
msk-donskaya-gspolyakov-gw-1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

```
Gateway of last resort is 198.51.100.1 to network 0.0.0.0
```

```
    10.0.0.0/8 is variably subnetted, 15 subnets, 4 masks
C       10.128.0.0/24 is directly connected, FastEthernet0/0.3
L       10.128.0.1/32 is directly connected, FastEthernet0/0.3
C       10.128.1.0/24 is directly connected, FastEthernet0/0.2
L       10.128.1.1/32 is directly connected, FastEthernet0/0.2
C       10.128.3.0/24 is directly connected, FastEthernet0/0.101
L       10.128.3.1/32 is directly connected, FastEthernet0/0.101
C       10.128.4.0/24 is directly connected, FastEthernet0/0.102
L       10.128.4.1/32 is directly connected, FastEthernet0/0.102
C       10.128.5.0/24 is directly connected, FastEthernet0/0.103
L       10.128.5.1/32 is directly connected, FastEthernet0/0.103
C       10.128.6.0/24 is directly connected, FastEthernet0/0.104
```

```
--More--
```

```
sch-sochi-gspolyakov-gw-1(config)#router ospf 1
sch-sochi-gspolyakov-gw-1(config-router)#
sch-sochi-gspolyakov-gw-1(config-router)#r
sch-sochi-gspolyakov-gw-1(config-router)#rou
sch-sochi-gspolyakov-gw-1(config-router)#router-id 10.128.254.4
sch-sochi-gspolyakov-gw-1(config-router)#net
sch-sochi-gspolyakov-gw-1(config-router)#network 10.0.0.0 0.255.255.255 area 0
sch-sochi-gspolyakov-gw-1(config-router)#exit
sch-sochi-gspolyakov-gw-1(config)#|
```

```
provider-gspolyakov-sw-1(config)#vlan 7
provider-gspolyakov-sw-1(config-vlan)#encap
provider-gspolyakov-sw-1(config-vlan)#encaps
provider-gspolyakov-sw-1(config-vlan)#encapsulation dot1Q 7
                                     ^
% Invalid input detected at '^' marker.

provider-gspolyakov-sw-1(config-vlan)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet
(401).

provider-gspolyakov-sw-1(config-vlan)#e
provider-gspolyakov-sw-1(config-vlan)#nam
provider-gspolyakov-sw-1(config-vlan)#name q42-sochi
provider-gspolyakov-sw-1(config-vlan)#exit
provider-gspolyakov-sw-1(config)#int
provider-gspolyakov-sw-1(config)#interface vla
provider-gspolyakov-sw-1(config)#interface vlan7
provider-gspolyakov-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan7, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan7, changed state to up

provider-gspolyakov-sw-1(config-if)#no sh
provider-gspolyakov-sw-1(config-if)#no shutdown
provider-gspolyakov-sw-1(config-if)#exit
```

```
msk-q42-gspolyakov-gw-1(config)#interface f0/1.7
msk-q42-gspolyakov-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/1.7, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.7, changed state to up

msk-q42-gspolyakov-gw-1(config-subif)#enc
msk-q42-gspolyakov-gw-1(config-subif)#encapsulation dot1Q 7
msk-q42-gspolyakov-gw-1(config-subif)#ip addre
msk-q42-gspolyakov-gw-1(config-subif)#ip address 10.128.255.9 255.255.255.252
% 10.128.255.8 overlaps with FastEthernet0/0
msk-q42-gspolyakov-gw-1(config-subif)#descrip
msk-q42-gspolyakov-gw-1(config-subif)#description sochi
msk-q42-gspolyakov-gw-1(config-subif)#exit
msk-q42-gspolyakov-gw-1(config)#|
```

```
sch-sochi-gspolyakov-sw-1(config)#vlan 7
sch-sochi-gspolyakov-sw-1(config-vlan)#name q42-sochi
sch-sochi-gspolyakov-sw-1(config-vlan)#exit
sch-sochi-gspolyakov-sw-1(config)#inter
sch-sochi-gspolyakov-sw-1(config)#interface vlan7
sch-sochi-gspolyakov-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan7, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan7, changed state to up

sch-sochi-gspolyakov-sw-1(config-if)#no sh
sch-sochi-gspolyakov-sw-1(config-if)#no shutdown
sch-sochi-gspolyakov-sw-1(config-if)#exit
sch-sochi-gspolyakov-sw-1(config)#|
```

```
msk-hostel-gspolyakov-gw-1(config)#router ospf 1
msk-hostel-gspolyakov-gw-1(config-router)#route
msk-hostel-gspolyakov-gw-1(config-router)#router-id 10.128.254.2
msk-hostel-gspolyakov-gw-1(config-router)#netw
msk-hostel-gspolyakov-gw-1(config-router)#network 0.255.255.255 area 0
                                     ^
% Invalid input detected at '^' marker.

msk-hostel-gspolyakov-gw-1(config-router)#network 10.0.0.0 0.255.255.255 area 0
msk-hostel-gspolyakov-gw-1(config-router)#exit
msk-hostel-gspolyakov-gw-1(config)#router ospf 1
msk-hostel-gspolyakov-gw-1(config-router)#router-id 10.128.254.3
msk-hostel-gspolyakov-gw-1(config-router)#Reload or use "clear ip ospf process" command, for this to take effect

exit
msk-hostel-gspolyakov-gw-1(config)#exit
msk-hostel-gspolyakov-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
clear ip ospf process
Reset ALL OSPF processes? [no]:

msk-hostel-gspolyakov-gw-1#
```